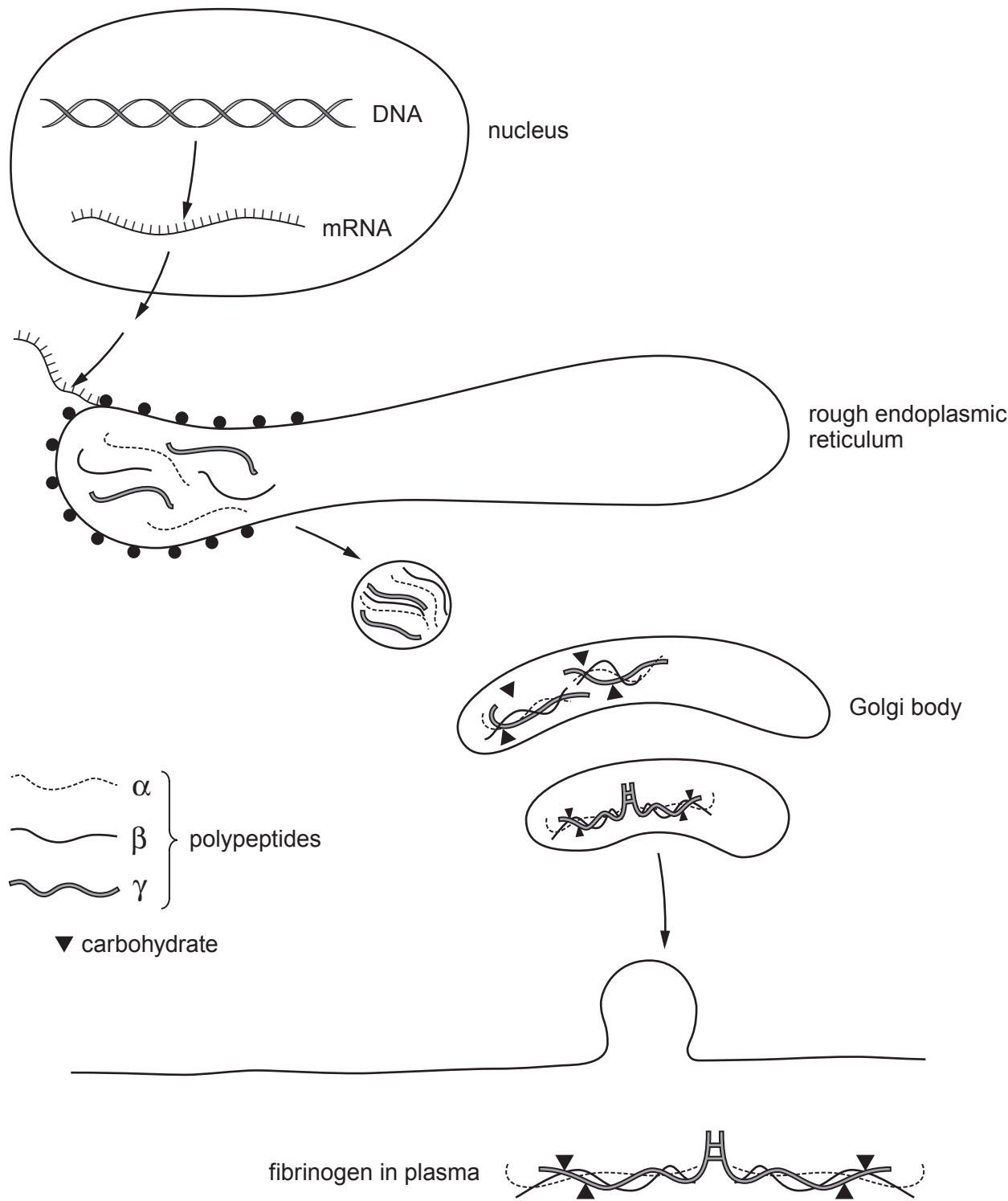


6. Fibrinogen is a soluble glycoprotein produced in liver cells and secreted into the plasma. It contains three types of polypeptide chain.

Image 6.1 shows the formation and secretion of fibrinogen.

Image 6.1



Examiner
only

(a) Use **Image 6.1** and your knowledge of proteins and protein synthesis to answer the following questions.

(i) Name the highest level of protein structure shown by fibrinogen. [1]

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(ii) Explain the role of the rough endoplasmic reticulum in the production of fibrinogen. [2]

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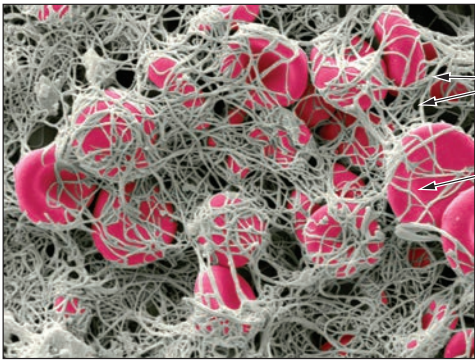
(iii) Suggest the role of the Golgi body in the production of fibrinogen. [2]

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- (b) Thrombin is an enzyme which is secreted into blood plasma when tissues are damaged. It catalyses the conversion of fibrinogen, in plasma, into the insoluble, fibrous protein, fibrin. Fibrin seals wounds in tissue. This is shown in **Image 6.2**.

Image 6.2



fibrin

red blood cell

Fibrin is a fibrous protein, fibrinogen is a globular protein.

Using the information provided, explain how the conversion of fibrinogen to fibrin allows the formation of a blood clot. [3]

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(c) Thrombin catalyses the hydrolysis of an arginine-glycine bond in fibrinogen as shown in **Image 6.3**.

Image 6.3

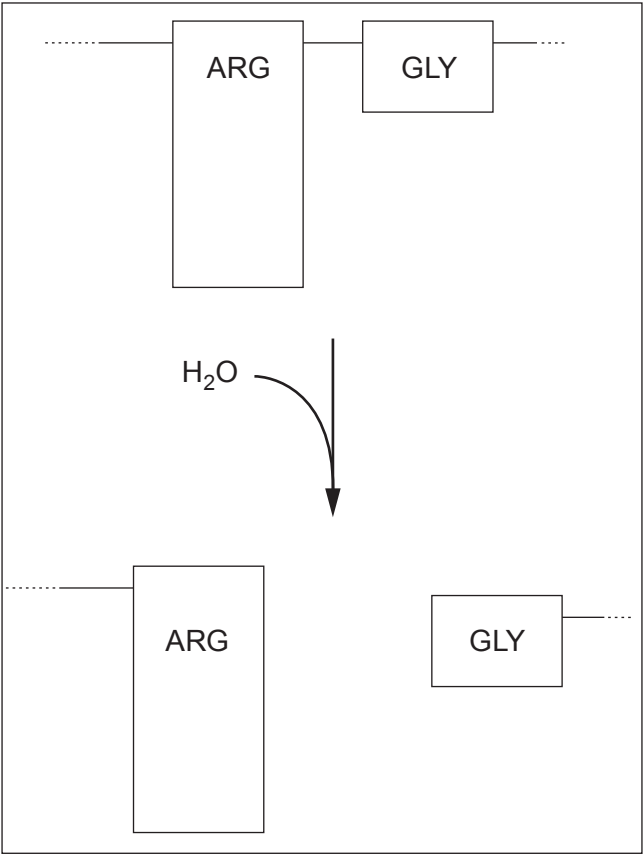
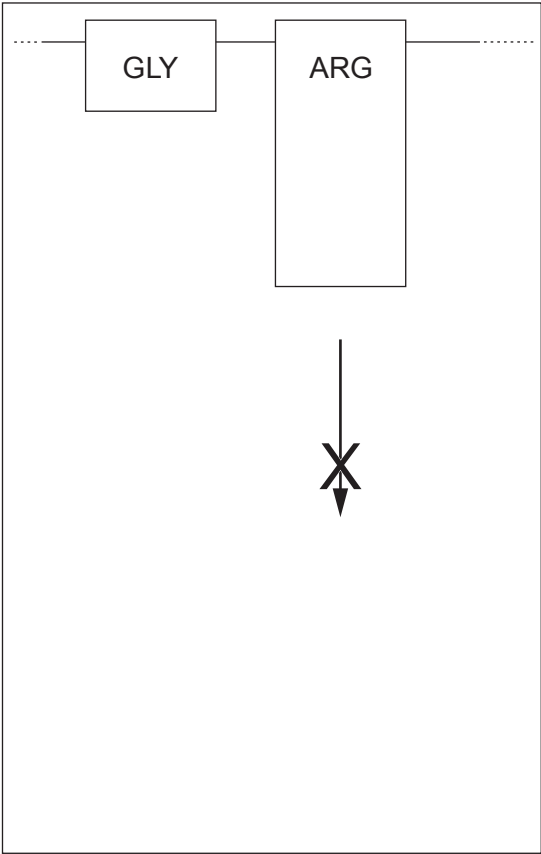


Image 6.4



Name the type of bond broken in **Image 6.3** and explain why thrombin can catalyse this reaction, but it would not catalyse the breaking of a glycine-arginine bond as shown in **Image 6.4**. [2]

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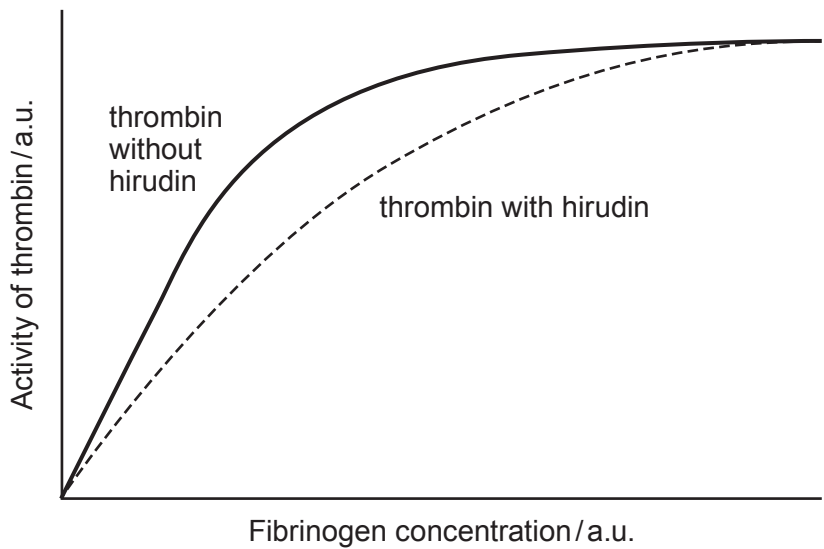
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- (d) Leeches are organisms which feed by sucking blood. Their saliva contains many components, including a protein called hirudin.

Graph 6.5 shows the inhibitory effect of hirudin on the activity of thrombin.

Graph 6.5



- (i) Using your knowledge of enzymes, identify and explain the type of inhibition shown by hirudin and suggest how hirudin enables the leech to feed effectively. [4]

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- (ii) Leeches for medical use are farmed in South Wales as a source of hirudin. Suggest a use for hirudin in medicine. [1]

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